



INDIAN INSTITUTE OF MANAGEMENT NAGPUR

POSTGRADUATE PROGRAMME (2025-27)

TERM IV

Data Driven Marketing

Project Final Report

Topic-

**Structural Category Decline and the Cross-Sell Opportunity in a
Multi-Branch Consumer Durables Dealership, Varanasi**

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EXECUTIVE SUMMARY

- **Corrected revenue base.** Cross-checking invoice versus line item level revenue for 67,334 lines revealed two discrepancies: the category revenue calculated from total invoice figures counted multi-category invoices twice (in four categories alone, there was ₹472.0m worth versus an actual FY2023-24 figure of ₹435.5m) and the FY2024-25 figure misses out about ₹56.7m of billing from February to March. Conclusion: the -13.7%/+9.3% V-shape is an illusion; the like-for-like erosion is 7.9%, from ₹364.5m to ₹335.7m.
- **Category and seasonality (A).** The only category that exceeds its FY2023-24 levels by any amount is AC (+16.8%), while all others have fallen by 10.2%-32.0%. April-June makes up 39%-49% of total revenue against a consistent 12% monsoon quarter. Conclusion: depletion of six long replacement cycle categories, not competitive inability.
- **Market basket — support, confidence, lift (B).** Of 36,056 invoices and 19,736 customer baskets, AC has lift below 1.0 versus every other category (0.475-0.849 invoice; 0.707-0.960 customer), while TV-Refrigerator-Washing Machines are grouped at 1.28-1.61. Conclusion: the sole growth channel creates isolated customers; consistency for three years precludes a timing artefact.
- **Value gap in cross-selling (B).** Customers buying more than one category are worth 3.4-3.7× those purchasing a single category, but the percentage of single category buyers rose 64.1% to 69.4%. Conclusion: 7,999 named customers in FY2025-26 compared to 6,886 two years ago, generating less revenue like-for-like due to declining value.
- **Seasonal pricing and RFM (C, D).** AC's price index coefficient of variation is 0.0072 – lower than Television

1. INTRODUCTION

A three-year period from a multi-branch Varanasi dealership dealing with LG white goods and Kent purifiers. The task was to identify a business problem evident within the organization itself through the use of its own internal information sources – analysis through observation, not technique through application. Four modules link up into a single argument: A determines what is going on and why, D to whom, B what could be done about it, C when and at what cost.

Context. Indian refrigerators have nine or ten year compressors warranties and washing-machine motors ten years or longer; televisions last long but get replaced by upgrades in six-and-a-half years or so. ACs have technical lives between ten and fifteen years, however, according to a 2025 iFOREST study 80% of units older than five years need to have their refrigerant replenished annually as opposed to one third of new units, making for premature replacement in year five. Varanasi reports June maximum temperatures of 44-47 degrees Celsius and January minimums of 9 degrees Celsius, while tourism post-Corridor development has increased hospitality purchases commercially.

2. DATA DESCRIPTION

The combination of three GST registers contained 67,341 rows (individual invoice lines), 36,056 invoices, and ₹1,220,481,886 billed. Rows consist of an invoice line including the invoice number and date, name of the customer, SKU, HSN, quantity, negotiated rate, taxable value, CGST/SGST/IGST, GST, and invoice-level gross amount. There is no unique key for the customer (names only) and there is no information about costs, margins, or promotions.

Seven duplicates removed (67,334 rows left). Invoice date processed using the day-first format and fiscal month generated. HSNs were categorized into seven groups, top 15 HSNs accounted for 96.6% of revenue. Three flags used without removing rows: zeros (3,372), only numerical SKU (187), and cash customers (4,342), where the latter was excluded from all customer analysis and left 19,736 customers. LINE_REVENUE (taxable value + all GST) accounts for 100.1% of the unique invoices' gross amount in all three years and forms the

3. QUESTIONS ASKED

- Q1. Is the trend legitimate, or is -13.7%/+9.3% just a mirage?
- Q2. Is the drop due to marketing problems or due to structure?
- Q3. Are items sold together, and is the expanding category pulling the portfolio?
- Q4. How much are multi-category shoppers worth, and what direction is the mix headed?
- Q5. Is pricing adjusted for the season, and how concentrated is the revenue?
- Q6. Does RFM work if 83.8% of customers only shop once?
- Q7. What other analysis can be done based on the above data?

4. LOGIC OF METHOD

Feasibility screening came first, hence no technique was tried if there is nothing to support it. Basket size is sufficient (1.86 line/invoice; 54% of multi-line FY2025-26 invoices include items in different categories), price variation real, repeat purchase rare (10.7-15.2%), promotional data unavailable. Six techniques have full support, five partial support, six – conjoint, market share simulation, advertising effectiveness and planning, promotions and display planning, network/viral – are infeasible without primary data.

Line sales, not total invoice amount. Category is line level, grand amount invoice level, thus assigning it per category would count multi-category invoices twice. This mistake was caught during the reconciliation process – when parts exceed the whole is wrong by definition – and has reversed the main finding, making AC the third largest decreasing category.

Like-for-like window. February FY2024-25 has 298 invoices and March 35, whereas in neighbouring periods there are about 800 invoices and ~1,100. A dealership does not go from 1,054 invoices in November to 35 in March and back to 1,249; this is a shortened export, therefore all comparisons are made April-January.

Support, confidence and lift altogether, at two levels. Support determines whether a rule appears often enough to take action, confidence conditional probability, lift correction of the consequent base rate. Confidence alone favors large categories, thus a rule may have high confidence but negative association. Invoice baskets (one visit) and customer baskets (three years of ownership) have been created since decade-cycle categories do not tend to buy each other in one basket; agreement across levels guarantees AC result validity.

Price normalisation, recalibrated RFM. Every rate is calculated in relation to its own product's three-year average before averaging monthly, thus preventing price changes from mimicking mix change. RFM's FMCG interpretation is rejected (Q6).

5. INFERENCES

Depletion, not a cycle. Like-for-like -1.3% then -6.6%; -7.9% over two years (Appendix B).

Concentration of structure and category. AC +16.8%, Television -24.2%, WM Automatic -32.0%, Water Purifier -20.4%, Refrigerator -16.0%: replacement cycles predict it. Six categories with ten-to-fifteen-year lifespan and warranty periods of nine to twenty years are depleting; AC, which sees its costs inflecting after year five and demand driven by climatic conditions, is the exception. Advertising in a depleting category means taking market share in a static pool, not in a growing pool.

Support and confidence validate association of lift measures rather than size. Refrigerator → Other displays the highest support (0.0902) and confidence (0.328) of any pair, but lift 0.856: "Other" is a large residual, appearing by chance and negative association. Television → WM Semi-Automatic shows less than half that support (0.0353) and roughly half that confidence (0.167) but maximum lift at 1.611: complementarity. Confidence alone would promote the wrong rule. Refrigerator → Television maximises both (support 0.0790 and lift 1.354) and is the anchor for §6.3.

Controllable failure follows AC purchase. AC is negatively associated with every category at both levels, while TV–Refrigerator–Washing Machine reaches 1.28–1.61 — household formation purchase bundle. AC occasion is an urgent, hot and narrow one; there is no bridge to portfolio. This is why AC grows by 16.8% while like-for-like revenues are down 7.9%.

Q6 — RFM is valid on two dimensions only. With 83.8% single-purchase and mean frequency 1.66, F is nearly degenerate: cannot be quintile-scored and banded, thus RFM is collapsed into R–M. Any churn analysis is invalid here: on decade-cycle goods, low frequency is satisfied customer, and the 2,965 customers with 874 days of inactivity are not churning but not yet due. Only value concentration and contactability remain valid: 351 Champions make 24.4% of revenue, top decile makes 46.1%. Degenerate F is itself the insight: in places where frequency is high (28.2 orders per customer, ₹806k average), it isolates commercial account within customer file. Ten-year period needed for true CLV.

Q7 — Scope for further analysis. Feasible now but out of scope: SKU-level assortment analysis and dead stock rationalization, as well as branch-level basket analysis. Available with minor additional data: margin-based CLV (cost of goods sold), price elasticity (\$6.4 experiment), and replacement-timing survival analysis (install dates from now).

6. APPLICATION

1. **Identity capture in point of sale** - prerequisite. Phone number in invoice of every sale including cash sales. Around 1,200–1,800 lines a year have no identity – without it nothing below is actionable and measurable.

2. **Convert the AC occasion into a portfolio relationship.** The biggest opportunity, as the only growing category is the least cross-sold. Make the AC installation appointment a touchpoint design - a home visit, which none of other categories generates, along with a voucher which can be redeemed later in July – September, thus dealing with seasonality from the beginning. Risk: The AC purchaser could be a tenant or a hospitality company and unresponsive; test the idea cheaply first.

3. **Off-lift targeting to next best category.** Allocate one of the 5,554 single-category purchasers an empirical next best category - Refrigerator owners → WM Semi-Automatic (1.599) and Television (1.347); Television owners → WM Semi-Automatic (1.588) and Automatic (1.477) – prior to the festive season in October.

4. **Test trough stimulation; do not repricing.** Vary effective price with bundling, warranty or installation services on certain high volume SKUs in July – September, relative to SKU levels of previous year. A dealer selling on trust and referral principle may find that discounting is more expensive than margin saved.

5. **Commercial customers management as accounts.** Make the 351 Champions as named accounts with pre-season contacts and stock commitments before April.

6. **Match working capital and staff capacity to the demand curve.** Given that 39-49% of revenue is made in April-June while 12% in monsoon quarter, flat resourcing is inefficient - margin increase requiring no change in customer behavior.

An idea to follow from findings: fast-moving electronics to drive the purchase frequency, which this business lacks. Not supported by current dataset, lacking any transactions of that sort. Appendix H documents this idea as a hypothesis for further research.

APPENDICES

Appendix A — Data Preparation Log

Step	Action	Rows affected	Reason
1	Unioned three fiscal-year registers	67,341 in	Joined on common columns; year-unique fields retained as optional attributes
2	Tagged FISCAL_YEAR	All	Identify source year for every row
3	Parsed INVOICEDATE day-first	All	Derived MONTH, FISCAL_MONTH (Apr=1..Mar=12)
4	Removed exact duplicate rows	7 removed	Byte-identical repeated lines; cleaned master = 67,334
5	Mapped HSNCODE to CATEGORY	All	Top 15 codes cover 96.6% of revenue; remainder grouped as Other
6	Computed LINE_REVENUE	All	CORRECTION: category is line-level, so revenue must be line-level
7	Flagged zero-value lines	3,372 flagged	GST rate nil — bundled/warranty items. Flagged, not deleted
8	Flagged bad SKU codes	187 flagged	Purely numeric PRODUCT field — mis-entered master data
9	Flagged cash customers	4,342 flagged	Anonymous walk-ins; excluded from all customer-level analysis
10	Derived BRANCH	All	Alphabetic prefix of invoice number
11	Built invoice-level table	36,056 invoices	Grand amount verified constant within each invoice
12	Identified FY2024-25 gap	See Appendix C	Feb and Mar materially incomplete; like-for-like window adopted

LINE_REVENUE reconciliation to distinct-invoice grand amount: 1.00105 (FY2023-24), 1.00083 (FY2024-25), 1.00074 (FY2025-26).

Appendix B — Analysis A: Category Revenue

Table B1 — Full-year reported basis (₹, LINE_REVENUE). Not comparable: see Appendix C.

Category	FY2023-24	FY2024-25	FY2025-26	YoY 24-25	YoY 25-26
Air Conditioner	139,511,416	149,978,183	166,097,409	+7.5%	+10.7%
Television	88,514,554	63,578,447	69,894,293	−28.2%	+9.9%
Refrigerator	79,051,398	63,930,849	68,593,837	−19.1%	+7.3%
Other	59,999,720	47,725,059	53,557,597	−20.5%	+12.2%
WM Automatic	43,531,605	31,283,790	30,656,039	−28.1%	−2.0%
WM Semi-Automatic	18,105,668	14,169,605	16,260,717	−21.7%	+14.8%
Water Purifier RO	6,742,720	4,999,671	5,369,000	−25.9%	+7.4%
TOTAL	435,457,081	375,665,606	410,428,892	−13.7%	+9.3%

Table B2 — Like-for-like April–January window (₹). This is the reliable view.

Category	FY2023-24	FY2024-25	FY2025-26	YoY 24-25	YoY 25-26	2-yr
Air Conditioner	116,949,119	143,419,756	136,574,334	+22.6%	−4.8%	+16.8%
Television	74,621,782	60,890,400	56,528,968	−18.4%	−7.2%	−24.2%
Refrigerator	65,403,492	61,576,644	54,932,988	−5.9%	−10.8%	−16.0%
Other	49,811,220	45,745,746	44,752,275	−8.2%	−2.2%	−10.2%
WM Automatic	36,853,118	29,468,022	25,067,175	−20.0%	−14.9%	−32.0%
WM Semi-Automatic	15,219,184	13,610,144	13,406,423	−10.6%	−1.5%	−11.9%
Water Purifier RO	5,600,618	4,901,071	4,456,150	−12.5%	−9.1%	−20.4%
TOTAL	364,458,534	359,611,784	335,718,314	−1.3%	−6.6%	−7.9%

Appendix C — Data Completeness Check and Seasonality

Table C1 — Invoice count by fiscal month. February and March FY2024-25 are the gap.

Year	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
FY2023-24	1,798	1,633	1,700	720	586	506	688	1,716	836	798	817	1,072
FY2024-25	2,202	1,835	1,488	649	447	386	1,339	1,054	668	687	298	35
FY2025-26	1,941	1,797	1,213	621	449	475	1,216	961	756	679	741	1,249

Feb + Mar revenue: ₹70,913,443 (FY2023-24), ₹16,053,821 (FY2024-25), ₹74,666,927 (FY2025-26). Estimated shortfall ₹56,736,364.

Table C2 — Total invoice revenue by month (₹) and average share of fiscal year.

Month	FY2023-24	FY2024-25	FY2025-26	Avg share
April	63,059,300	75,661,637	69,428,278	17.2%
May	54,451,840	58,669,988	62,755,311	14.5%
June	52,030,697	48,200,762	38,490,831	11.4%
July	22,751,512	20,337,821	19,476,755	5.1%
August	17,108,915	14,516,306	13,153,529	3.7%

September	15,867,378	12,994,394	16,482,623	3.7%
October	25,399,518	44,926,531	37,537,982	9.0%
November	56,408,527	39,777,831	33,412,278	10.6%
December	26,062,219	19,232,996	20,246,689	5.3%
January	30,947,324	24,982,003	24,475,920	6.6%
February	31,791,762	13,920,569	30,310,492	6.1%
March	39,121,681	2,133,252	44,356,435	7.1%

April–June contributes 39–49% of annual revenue; the July–September monsoon quarter a stable 12.0–12.8% in every year.

Appendix D — Analysis B: Market Basket, Support / Confidence / Lift

Table D1 — Invoice basket (36,056 transactions), ranked by lift.

Category A	Category B	Pairs	Support	Conf. A→B	Lift
Television	WM Semi-Automatic	1,274	0.0353	0.167	1.611
Refrigerator	WM Semi-Automatic	1,588	0.0440	0.160	1.547
Television	WM Automatic	1,322	0.0367	0.173	1.523
Television	Refrigerator	2,848	0.0790	0.373	1.354
Refrigerator	WM Automatic	1,446	0.0401	0.146	1.284
WM Automatic	Water Purifier RO	132	0.0037	0.032	1.077
Other	WM Automatic	1,600	0.0444	0.116	1.021
Television	Other	2,931	0.0813	0.383	1.001
Other	WM Semi-Automatic	1,390	0.0386	0.101	0.973
Refrigerator	Other	3,253	0.0902	0.328	0.856
Air Conditioner	WM Automatic	989	0.0274	0.096	0.849
Other	Water Purifier RO	343	0.0095	0.025	0.829
Air Conditioner	Television	1,523	0.0422	0.148	0.700
Television	Water Purifier RO	147	0.0041	0.019	0.642
Air Conditioner	Other	2,414	0.0670	0.235	0.615
Refrigerator	Water Purifier RO	173	0.0048	0.017	0.582
Air Conditioner	Refrigerator	1,620	0.0449	0.158	0.574
Air Conditioner	WM Semi-Automatic	570	0.0158	0.056	0.537
Air Conditioner	Water Purifier RO	146	0.0040	0.014	0.475
WM Semi-Automatic	Water Purifier RO	45	0.0012	0.012	0.403
WM Automatic	WM Semi-Automatic	12	0.0003	0.003	0.028

Table D2 — Customer basket (19,736 customers, three-year portfolio ownership), ranked by lift.

Category A	Category B	Pairs	Support	Conf. A→B	Lift
Refrigerator	WM Semi-Automatic	1,546	0.0783	0.221	1.599
Television	WM Semi-Automatic	1,305	0.0661	0.220	1.588
Television	WM Automatic	1,431	0.0725	0.241	1.477
Other	WM Semi-Automatic	1,409	0.0714	0.191	1.378
WM Automatic	Water Purifier RO	163	0.0083	0.051	1.378
Other	WM Automatic	1,640	0.0831	0.222	1.360
Refrigerator	WM Automatic	1,541	0.0781	0.221	1.352
Television	Refrigerator	2,830	0.1434	0.477	1.347
Television	Other	2,969	0.1504	0.500	1.336
Other	Water Purifier RO	362	0.0183	0.049	1.334
Refrigerator	Other	3,207	0.1625	0.459	1.226
Air Conditioner	WM Automatic	1,135	0.0575	0.157	0.960
Refrigerator	Water Purifier RO	236	0.0120	0.034	0.920
Air Conditioner	Other	2,477	0.1255	0.342	0.913
Television	Water Purifier RO	196	0.0099	0.033	0.899
Air Conditioner	Water Purifier RO	225	0.0114	0.031	0.845
Air Conditioner	Television	1,767	0.0895	0.244	0.811
WM Semi-Automatic	Water Purifier RO	75	0.0038	0.027	0.748
Air Conditioner	Refrigerator	1,883	0.0954	0.260	0.734
Air Conditioner	WM Semi-Automatic	709	0.0359	0.098	0.707
WM Automatic	WM Semi-Automatic	130	0.0066	0.040	0.292

Air Conditioner is below 1.0 against every category at both levels. The lowest lift in the matrix (WM Automatic × WM Semi-Automatic, 0.028 / 0.292) confirms the two washing-machine variants are substitutes and must not be cross-offered.

Appendix E — Analysis B: Single vs Multi-Category Customer Value

Fiscal year	Customer type	Customers	Avg spend (₹)	Total revenue (₹)	Multi/single ratio	Single-cat share
FY2023-24	Multi-category	2,474	111,253	275,240,885	3.60	64.1%
FY2023-24	Single-category	4,412	30,940	136,507,808		
FY2024-25	Multi-category	2,049	114,012	233,610,690	3.67	65.7%
FY2024-25	Single-category	3,917	31,049	121,620,096		
FY2025-26	Multi-category	2,445	96,387	235,665,372	3.36	69.4%
FY2025-26	Single-category	5,554	28,728	159,553,681		

Appendix F — Analysis C: Product-Normalised Monthly Price Index

Category	Apr	Jun	Aug	Oct	Dec	Feb	Coeff. of variation
Television	1.0065	1.0169	1.0165	0.9883	0.9884	0.9983	0.0128
WM Semi-Automatic	1.0080	1.0251	1.0153	0.9928	0.9948	0.9959	0.0118
Water Purifier RO	0.9953	0.9870	1.0081	0.9999	1.0046	0.9895	0.0080
Air Conditioner	0.9941	1.0101	0.9919	1.0054	0.9940	1.0117	0.0072
Other	1.0005	1.0037	0.9930	0.9933	1.0064	0.9906	0.0068
Refrigerator	1.0027	1.0085	1.0000	0.9949	0.9951	0.9912	0.0062
WM Automatic	1.0050	1.0127	1.0017	0.9954	0.9981	0.9965	0.0055

Alternate months shown for space; the full twelve-month index is in the workbook (ANALYSIS_C_PRICING). Caveat: these are negotiated transaction prices, not experimentally controlled prices, so the evidence is correlational.

Appendix G — Analysis D: RFM Segmentation

Table G1 — Segment summary. Cash customers excluded; 19,736 named customers.

Segment	Customers	Revenue (₹)	Avg monetary (₹)	Avg freq.	Avg recency (days)	Rev. share
Potential Loyalist	7,536	303,475,328	40,270	1.11	201	26.11%
Champions	351	283,081,361	806,500	28.21	387	24.36%
At Risk (high value)	2,965	225,041,912	75,899	1.25	874	19.36%
Needs Attention	3,688	167,008,032	45,284	1.15	546	14.37%
Lapsed / Not Yet Due	4,667	109,707,367	23,507	1.04	885	9.44%
Loyal	529	73,884,532	139,668	3.02	512	6.36%
TOTAL	19,736	1,162,198,532	58,887	1.66	—	100%

Table G2 — Revenue concentration and frequency structure.

Metric	Value
Customers transacting once in three years	16,541 (83.8%)
Top 10% of customers — share of revenue	46.1%
Top 20% of customers — share of revenue	59.9%
Median recency	526 days
Mean frequency	1.66 invoices per customer

Champions average 28.21 invoices over three years — far outside any household replacement pattern. Combined with a parallel invoice series billed to trading names, this indicates the segment is substantially commercial and sub-dealer, which is why Recommendation 5 applies account-management rather than consumer methods.

Appendix H — Limitations

- 1. Observation window far shorter than the replacement cycle.** Three years against ten-to-fifteen-year cycles — between one fifth and one third of one cycle. The data structurally cannot distinguish a lost customer from one not yet due. Every retention and lifetime-value statement is bounded by this.
- 2. FY2024-25 completeness gap.** Mitigated by the April–January window, but the mitigation discards two genuine months from the other years and assumes the gap is confined to Feb–Mar, which cannot be verified from within the data.
- 3. No customer identifier.** Name-string matching both splits customers across spelling variants and merges distinct customers sharing a name. All customer counts carry unquantified error. A further 4,342 cash rows are excluded entirely.
- 4. No cost or margin data.** A sales register, not a P&L. Every value statement is revenue-only. A cross-sell that raises revenue could in principle reduce profit; recommendations should be re-tested against margin before significant spend.

- 5. Effectively no promotional data.** Discount, cashback and freight fields are populated in under 1% of rows. It cannot be resolved whether the firm genuinely does not discount, or discounts informally through negotiated basic rates — if the latter, Analysis C measures recorded rather than realised price.
- 6. Pricing evidence is correlational.** Observed variation confounds pricing decisions with negotiation, salesperson discretion, product age and mix. The finding that AC prices do not move seasonally is robust; the inference that moving them would raise revenue is not, which is why Recommendation 4 is an experiment.
- 7. Replacement-cycle explanation is directional, not proven.** The data holds no product age or installed-base information and cannot mark a purchase as first or replacement. Competitive entry, credit availability and e-commerce substitution cannot be excluded.
- 8. No external or market context.** Own sales only. It cannot be determined whether the 7.9% decline reflects a shrinking market in which the firm holds share, or a growing one in which it loses share — opposite strategic implications.
- 9. Branch effects excluded.** Two branches ceased trading and one opened across the window. Deliberately excluded as a network-strategy rather than marketing question, but the aggregate trend contains this unmodelled effect.
- 10. Category mapping is approximate.** HSN-based; the residual "Other" aggregates 76 codes spanning accessories, spares and installation materials and is not a coherent commercial category.
- 11. Fast-moving electronics adjacency is untested.** The hypothesis that a sub-five-year-cycle category would generate the purchase frequency this business lacks follows from the findings, but this dataset contains no such transactions. It requires its own feasibility study and is recorded as an observation, not a recommendation.

Appendix I — Submitted Analysis File: Excel Workbook

The complete analysis workbook **LG_Kent_Marketing_Analytics_FY2324_FY2526.xlsx** is submitted alongside this report in the same folder. Every figure quoted in the body and appendices is reproducible from it: the cleaned line-item dataset is carried in full, so the instructor can re-run any analysis from source without reference to external files. Sheet contents are as follows.

Sheet	Contents	Size
README	Business context, key verified numbers, contents guide	—
DATA_DICTIONARY	Every column defined; original vs derived fields	22 fields
PREPROCESSING_LOG	Every cleaning step in order, rows affected, reason	12 steps
DATA_GAP_CHECK	FY2024-25 Feb–Mar completeness test — read before any YoY comparison	—
ANALYSIS_A_CATEGORY	Category revenue by year, full-year basis, with reconciliation	Table B1
ANALYSIS_A2_LIKE_FOR_LIKE	Category revenue, comparable Apr–Jan window — the reliable view	Table B2
ANALYSIS_A_SEASONALITY	Monthly revenue, invoice counts, category revenue by month	Table C2
ANALYSIS_B_BASKET	Support, confidence and lift for every pair, both basket levels	Tables D1–D2
ANALYSIS_B_CROSSELL	Single vs multi-category customers and the value gap	Table E
ANALYSIS_C_PRICING	Product-normalised monthly price index, full 12 months, CV per category	Table F
ANALYSIS_D_RFM	RFM segmentation, segment sizes, revenue concentration	Tables G1–G2
KEY_FINDINGS	Numbered findings with supporting numbers	—
RECOMMENDATIONS	Prioritised recommendations with rationale and risk	—
LIMITATIONS	What the dataset cannot support	Appendix H
CLEANED_MASTER_DATA	Full cleaned line-item dataset with all derived columns and flags	67,334 rows
INVOICE_LEVEL	One row per invoice	36,056 rows

To reproduce: open CLEANED_MASTER_DATA and pivot on CATEGORY × FISCAL_YEAR using LINE_REVENUE (never GRANDAMOUNT, which is invoice-level — see Appendix A step 6). Restrict FISCAL_MONTH to 1–10 for the like-for-like view.